

Class Test 3

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Electricity and Electronics EBN 111

28 June 2021

Test Information

Maximum marks:	16	Full marks:	15
Duration of test:	35 min	Open/Closed book:	Open
Total number of pages (this page included)		9	

Lecturers: Dr D. le Roux, Dr S. Smith, Dr X. Ye**Assistant Lecturers:** M. Maritz, J. Thiselton, M. Trivedi, D. Wepener, L. Kokot**Important**

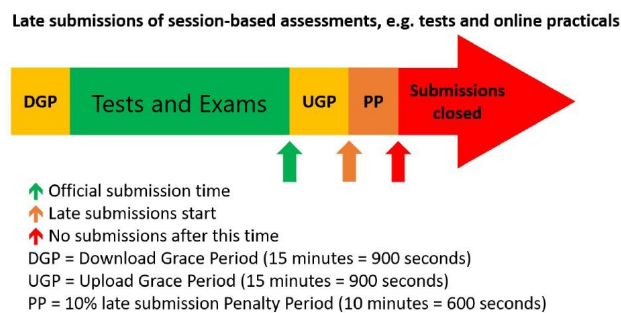
1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS Excel spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By submitting your class test, you declare the following:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test. Students must work through the test at an average rate of at least 2 minutes per mark.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions are accepted after the PP.

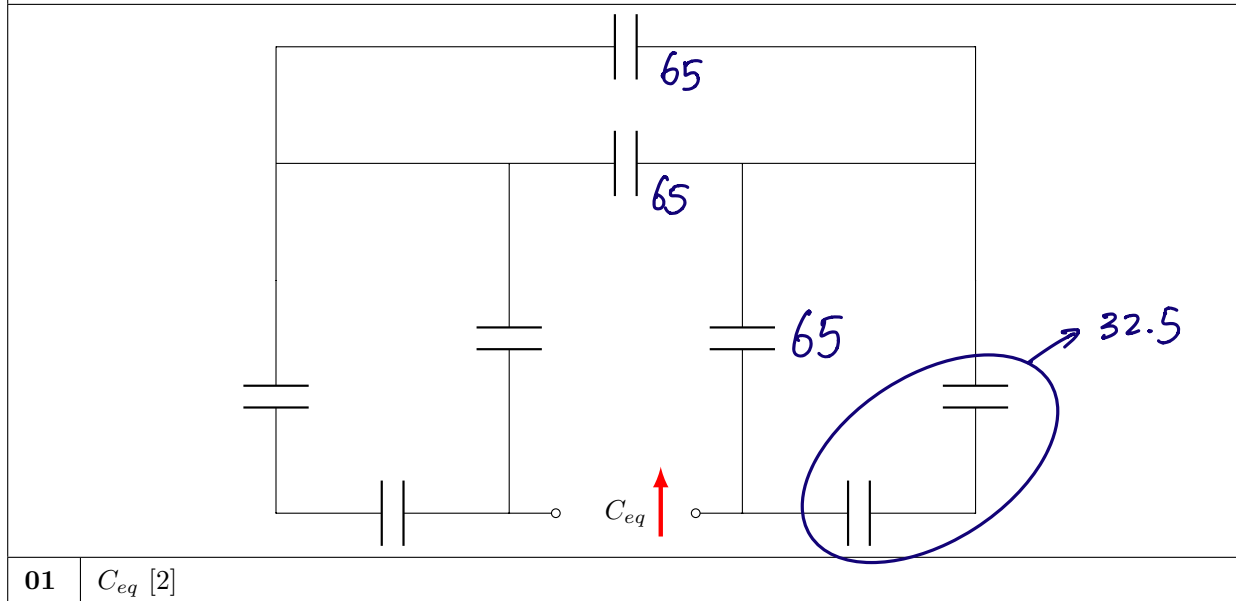


Additional guide for Complex numbers

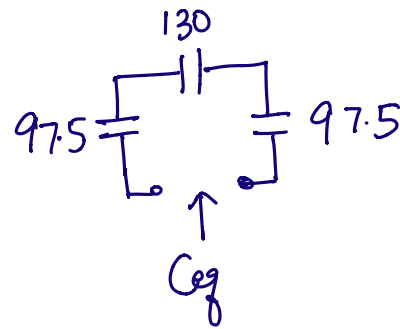
The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \deg$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

Question 1 [2]

Given: $C = 65$ Find C_{eq} in the circuit if the capacitance of **each capacitor** is C mF.01 C_{eq} [2]

$$\begin{aligned}
 C_{eq} &= \frac{130 \cdot \frac{97.5}{2}}{130 + \frac{97.5}{2}} \\
 &= \frac{130 \times 97.5}{260 + 97.5} \\
 &= 35.455 \text{ mF}
 \end{aligned}$$



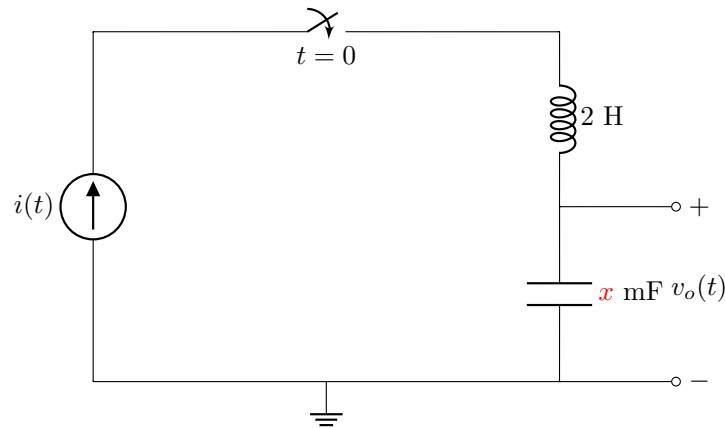
Question 2 [2]

$$x = 3.2$$

$$k = 9$$

Given: x, k .

In the circuit below, the inductor and capacitor are initially uncharged as the switch will only be closed at $t = 0$. Find $v_o(t)$ at $t = 2$ ms when $i(t) = 2t + k$ A.

02 $v_o(2 \text{ ms})$ [2]

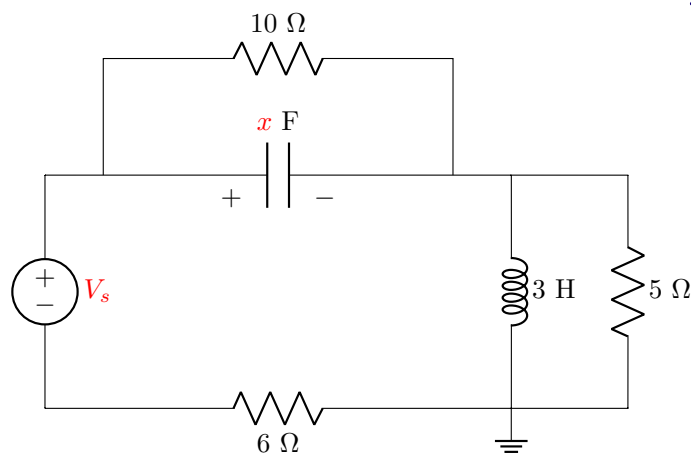
$$\begin{aligned}
 v_o &= \frac{1}{3.2 \times 10^{-3}} \int_0^t (2t + 9) dt \\
 &= \frac{1000}{3.2} [t^2 + 9t]_{0}^{2 \times 10^{-3}} \\
 &= \frac{1000}{3.2} [(2 \times 10^{-3})^2 + 9 \times 2 \times 10^{-3}] \\
 &= \frac{4 \times 10^{-3} + 18}{3.2} = 5.6263 \text{ V.}
 \end{aligned}$$

Question 3 [2]Given: V_s , x .

$$V_s = 64$$

Under steady DC conditions, find the energy w_C stored in the capacitor.

$$x = 20$$

**03** w_C [2]

$$V_C = 64 \times \frac{10}{10+6} = 40\text{ V.}$$

$$w_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 20 \times 40^2 = 16\text{ kJ}$$

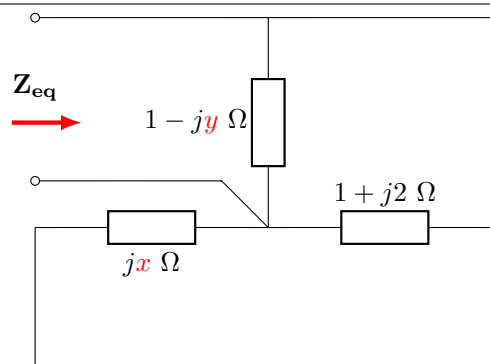
Question 4 [2]	
Given: a, b . $a=2, b=100$	
Find $\mathbf{I}_1$ in rectangular form.	
$\begin{bmatrix} a+j3 & -j2 \\ -j2 & 8-j5 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} j6 \\ b \end{bmatrix}.$	
04	$\mathbf{I}_1$ (rectangular form). [2]

$$\begin{aligned}
 \mathbf{I}_1 &= \frac{\begin{vmatrix} j6 & -j2 \\ 100 & 8-j5 \end{vmatrix}}{\begin{vmatrix} 2+j3 & -j2 \\ -j2 & 8-j5 \end{vmatrix}} = \frac{j6(8-j5) + j200}{(2+j3)(8-j5) + 4} \\
 &= \frac{30 + j248}{35 + j14} \\
 &= 3.1823 + j5.8128
 \end{aligned}$$

Question 5 [2]		$x=4$
Given: x, y .		$y=10$
A series RLC circuit has a resistor of $R = 80 \Omega$, an inductor of $L=x$ mH, and a capacitor of $C=y$ mF. If the circuit is connected to a voltage source of $v(t) = 1000\cos(2t)$ V, find the current $\mathbf{I}$ flowing through the circuit. Assume the current I exits the positive terminal of the voltage source. Provide your answer $\mathbf{I} = r\angle\theta$ A in polar form , where $r \geq 0$ and $-180^\circ \leq \theta \leq 180^\circ$.		
05	$\mathbf{I}$ (polar form). [2]	

$$\begin{aligned}
 I &= \frac{1000\angle 0^\circ}{80 + j2 \times 4 \times 10^{-3} - j \frac{1}{2 \times 10 \times 10^{-3}}} \\
 &= \frac{1000}{80 + j0.008 - j50} = 10.600\angle -32.001^\circ \text{ A}
 \end{aligned}$$

Question 6 [2]

Given: x , y .Find $\mathbf{Z}_{\text{eq}} = r \angle \theta$ in **polar form**, where $r \geq 0$ and $-180^\circ \leq \theta \leq 180^\circ$.

$$x = 30$$

$$y = 8$$

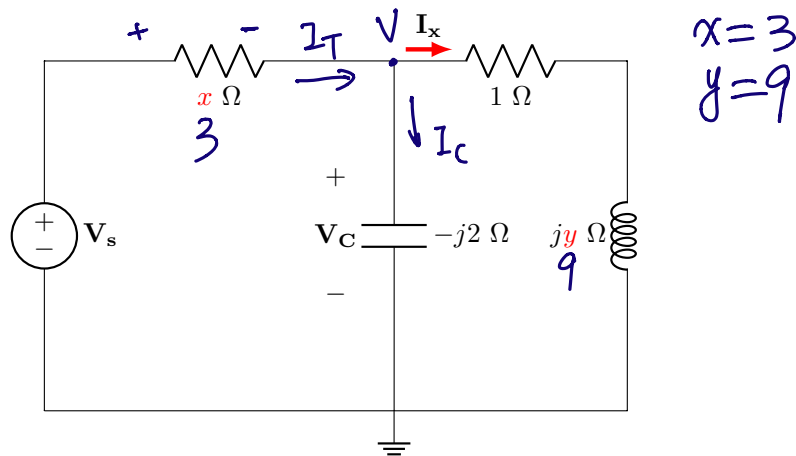
06 $\mathbf{Z}_{\text{eq}}$ (polar form). [2]

$$Z_1 = j30 \parallel 1 + j2 = \frac{j30(1 + j2)}{1 + j32} = \frac{36}{41} + j\frac{78}{41} \Omega$$

$$Z_{\text{eq}} = Z_1 \parallel (1 - j8) = \frac{(1 - j8) \times (\frac{36}{41} + j\frac{78}{41})}{1 - j8 + \frac{36}{41} + j\frac{78}{41}}$$

$$= 2.6477 \angle 55.231^\circ \Omega$$

Questions 7 to 8 [4]

Given: x, y .07 Find V_s in rectangular form if $I_x = 3\angle 90^\circ$ A. [2]08 Find V_C in rectangular form if $V_s = 10\angle 0^\circ$ V. [2]

$$07: V = j3(1+j9) = -27 + j3$$

$$I_c = \frac{-27 + j3}{-j2} = -1.5 - j13.5$$

$$\frac{V_s - V}{3} = I_T = j3 + I_c = -1.5 - j10.5$$

$$\Rightarrow V_s = 3(-1.5 - j10.5) + (-27 + j3) \\ = -31.5 - j28.5 \text{ V}$$

$$08: Z = -j2 \parallel (1+j9) = \frac{-j2(1+j9)}{-j2 + 1+j9} = 0.08 - j2.56$$

$$V_c = \frac{10Z}{x+Z} = \frac{0.08 - j2.56}{3.08 - j2.56} = 4.2394 - j4.7880 \text{ V}$$

Grand Total : [16]